

PCB DESIGN TASK

Fixed Bias Bistable Multivibrator

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1. Design Task

Design and create a complete PCB layout for a Fixed Bias Bistable Multivibrator using PCB design software. The task includes schematic capture, component footprint assignment, PCB placement, routing of PCB tracks, design-rule checking (DRC), and generation of manufacturing-ready Gerber files.

2. Objective

- To design a working fixed-bias bistable multivibrator circuit using transistor switching.
- To convert the circuit schematic into a practical single-sided or two-layer PCB layout.
- To place components according to electrical and mechanical requirements and route reliable PCB tracks.

To verify the PCB using electrical rules and design-rule checks.

- To generate Gerber and drill files suitable for PCB fabrication.

3. Suggested Circuit / Components

Component	Suggested value	Quantity
NPN transistor	BC547 / 2N2222	2
Collector resistor	4.7 k Ω	2
Base resistor	47 k Ω	2
Cross-coupling capacitor	10 μ F	2
Emitter resistor	1 k Ω	2
Supply	5–12 V DC	1
LED (optional)	5 mm	2
LED resistor (optional)	1 k Ω	2
Terminal / connector	2-pin	1

4. PCB Design Procedure

Step 1 – Schematic Capture: Draw the fixed-bias bistable multivibrator circuit in KiCad, Altium Designer, EasyEDA, Eagle, or similar PCB software. Label VCC, GND, and output/test points clearly.

Step 2 – Footprint Assignment: Assign through-hole or SMD footprints to every component. Confirm transistor pin numbering and capacitor polarity before proceeding.

Step 3 – Netlist / PCB Update: Transfer the schematic design to the PCB editor and verify that all electrical nets are correctly imported.

Step 4 – Board Outline: Create a compact rectangular PCB outline with suitable mounting and connector locations.

Step 5 – Component Placement: Place the transistors and cross-coupling components close together. Keep power and ground connections short and provide accessible test points.

Step 6 – Track Routing: Route signal and bias tracks first, followed by power and ground. Use suitable track widths and avoid unnecessary sharp corners or long parallel runs.

Step 7 – Ground / Power Integrity: Use a clear ground return path. Keep supply wiring short and place decoupling close to the active devices.

Step 8 – Design Rule Check: Run DRC to identify clearance, unconnected-net, footprint, and board-edge violations. Correct all critical errors.

Step 9 – PCB Verification: Inspect the final board in 2D and 3D views and confirm component orientation, reference designators, polarity marks, and connector accessibility.

Step 10 – Gerber Generation: Generate copper, solder-mask, silkscreen, board-outline, and drill files using the PCB software's manufacturing-output tool.

5. Recommended PCB Layout Guidelines

- Use a practical track width such as 0.25–0.50 mm for low-current signal tracks, subject to the PCB manufacturer's capabilities.
- Maintain adequate track-to-track and track-to-board-edge clearance according to the selected fabrication rules.
- Place polarized capacitors and transistors with clearly marked orientation.
- Keep cross-coupling connections short to reduce unwanted noise and parasitic effects.
- Provide clearly labeled VCC, GND, and output test points.
- Add reference designators and polarity indicators to the silkscreen.
- Check the transistor datasheet because BC547/2N2222 package pinouts can differ between manufacturers.

6. Required Gerber / Manufacturing Files

- Top copper (F.Cu) Gerber
- Bottom copper (B.Cu), if a two-layer PCB is used
- Top solder mask
- Bottom solder mask, if applicable
- Top silkscreen
- Bottom silkscreen, if applicable
- Board outline / Edge.Cuts
- Excellon drill file
- Fabrication drawing or drill map, if required

7. Software Options

The design may be implemented using KiCad, EasyEDA, Autodesk Fusion Electronics/Eagle, Altium Designer, or another PCB CAD package. The exact menu names and Gerber export settings depend on the selected software.

8. Expected Output

- Completed electrical schematic of the Fixed Bias Bistable Multivibrator.
- PCB layout showing component placement and routed tracks.
- DRC report with critical violations resolved.
- 2D/3D PCB preview for visual inspection.
- Complete Gerber and drill files packaged for fabrication.
- A short design report documenting component values, board dimensions, routing rules, and verification results.

9. Design Verification

Before fabrication, verify circuit connectivity against the schematic, transistor pin configuration, capacitor polarity, power-rail continuity, track clearances, unconnected nets, and board-edge clearance. After fabrication, visually inspect the PCB and test the bistable switching action using a suitable DC supply and oscilloscope or logic probe.

10. Final Deliverables Checklist

Deliverable	Status
Schematic file	I
PCB layout file	I
DRC report	I
3D/2D layout image	I
Gerber files	I
Drill file	I
Final design report	I

Note: This document is a design-task specification. Actual Gerber files must be generated from the completed PCB design in the selected PCB CAD software; they cannot be produced reliably without a finalized schematic, footprints, board dimensions, and routing.